

# Jiaxin (Jeremiah) Pan

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Portfolio · GitHub · LinkedIn

## Research Profile

Physics undergraduate working on AI for Science, generative sampling, interpretable materials representations, and language-model reasoning. Research interests connect statistical mechanics with reproducible scientific computing and reliable machine-learning methods.

## Education

2023–2027 **B.Sc. in Physics (with honours) — Expected**

The Chinese University of Hong Kong, Shenzhen · Shenzhen, China

GPA: 3.61/4.0. Four-year honours degree with a required research thesis. Admitted to Economics in 2023; completed Year 1 foundation courses before transferring to the B.Sc. Physics programme in 2024 to pursue my passion for sciences.

## Exchange and Visiting Programmes

2026 (Spring) **Visiting Student**

University of Auckland · Auckland, New Zealand

Nominated by CUHK-Shenzhen; completed advanced coursework in Physics, Computer Science, and AI, including COMPSCI 361 and COMPSCI 220.

2025  
(Summer) **Summer Visiting Student**

University of California, Berkeley Extension · Berkeley, CA, USA

Completed coursework in physics and neuroscience; developed cross-cultural academic communication and independent learning skills.

## Ongoing Research

Ongoing **Data-Free Generative Sampling of the Two-Dimensional XY Model**

Undergraduate Researcher, CUHK-Shenzhen

Supervised by Prof. Kai Zhou. Provisional portfolio title, not a finalized thesis title.

- Contributed to adapting Stochastic Path Sampling (SPS) to periodic XY angle variables. Training uses the XY energy and forward/backward path probabilities without pre-generated equilibrium configurations as training data. Generated paths are evaluated with path-space independence Metropolis–Hastings (MH) correction.
- Earlier exploration used supervised Flow Matching trained on Wolff-generated configurations. Wolff sampling is an independent validation reference for the current method.
- Completed a training-budget comparison across three seeds, 5,000/10,000/15,000 parameter updates, and 32/64/128 generation steps. Python/PyTorch GPU experiments run through Slurm.
- Evaluated energy, vortex counts, magnetization, spatial correlations, acceptance, effective sample size per sampling second, and consecutive rejection streaks.
- Estimated energy sampling efficiency increased in all nine seed/step settings from 5,000 to 10,000 updates; from 10,000 to 15,000, six improved and three declined. All nine settings at both later checkpoints passed predefined physical-statistics screening criteria; this supports, but does not prove, complete equilibration.
- High relative path weights can cause prolonged MH rejection streaks. Acceptance alone does not determine sampling efficiency. Analysis and report preparation are underway.

The limited  $4\times 4$  lattice study at one temperature near the expected Berezinskii–Kosterlitz–Thouless (BKT) transition does not establish thermodynamic critical behavior, a solution to critical slowing down, universal convergence, or an advantage over Wolff. SPS is an existing method.

## Ongoing **Interpretable Crystal Representations for AI-Assisted Materials Design**

HSSRLM — AI for Science — Working paper

An initial AAAI submission plan was deferred while another manuscript took priority. Possible future venues include IJCAI and ACL, subject to fit and further development. A collaborator indicated that my current contributions could support co-first authorship; co-first, second, or other authorship positions depend on final contributions and author agreement. Submission plans and final author order remain unconfirmed. Framework: compact hierarchical descriptors form an interpretable interface between natural language and porous materials, including metal–organic frameworks (MOFs) and zeolites.

- Hierarchical Structure-Sequence Representation (HSSR) organizes composition, lattice parameters, symmetry, topology, building units, and selected pore-related properties into structured text. The repository implements T5 (Text-to-Text Transfer Transformer) infrastructure for bidirectional text/descriptor translation.
- Contributed to evidence-aware Crystallographic Information File (CIF)-to-descriptor preprocessing that separates known, inferred, and missing information, preserves provenance, and routes incomplete records for review.
- Contributed to multi-candidate screening with descriptor validation, node/linker/topology connectivity checks, explicit failure reporting, and descriptor-level duplicate tracking.
- Contributed to reproducible structure-equivalence audits on three publicly sourced MOF structures, examining equivalent representations and information lost by coarse descriptors. This is a limited study, not evidence of universal representation invariance.

Framework implementation and targeted reliability audits are complete; broader dataset validation and model evaluation remain in progress. HSSR omits complete atomic coordinates and a full bonding graph, so its outputs are descriptors rather than complete atomistic CIF structures. Verification establishes compatibility with implemented rules, not physical validity or synthesizability. CrysSpec (Crystal Specification) is a provisional working name. The framework also includes a 26-record metal-node registry that addresses ambiguous aliases and preserves source material for continued chemical validation.

## **Additional Research Experience**

### 2026 (Spring) **Remote Research Assistant**

INSAIT · Bulgaria / Remote  
Under Dr. Jinjin Gu.

- Analysed large language model reasoning and post-training, including Surgical Post-Training (SPOT), implicit regularization, minimal-edit data rectification, and catastrophic forgetting.
- Studied the trade-off between reasoning performance and foundational knowledge retention; prepared research syntheses and contributed to seminars.

### 2025 (Fall) **Guided Research on Van der Waals Integration and 2D Electronics**

CUHK-Shenzhen, Prof. Qi Qian's Lab

Participated in fabrication and characterization of MoS<sub>2</sub> transistors through van der Waals integration and dry transfer of 2D heterostructures. Studied metal–semiconductor contacts and polarity modulation; gained hands-on nanofabrication and electrical characterization experience, conducted literature review, and prepared an end-of-term report.

### Earlier work **Guided Research Training in Generative AI and Diffusion Models**

CUHK-Shenzhen, Prof. Kai Zhou's Group

Earlier training in score-based generative modelling and physical dynamics.

- Explored connections between stochastic differential equations and diffusion models, including the denoising score-matching objective.
- Developed simulations of a one-dimensional double-well potential and visualized generative trajectories.
- Analysed learned score fields and local stability using Jacobian matrices; compared Langevin-sampled steady-state distributions with the Gibbs distribution.

## **Conference**

### 31 Jan 2026 **CUHK-Shenzhen Undergraduate Research Conference (URC2026)**

Second Undergraduate Research Conference · Shenzhen, China

Presented “Self-Confidence in Math and STEM Major Choice at CUHK-Shenzhen” at my first research conference. The study examined the relationship between mathematical self-confidence and STEM major choice. [Research paper \(public copy\)](#); presentation certificate and on-site photographs on the portfolio.

## Course Projects

- 2025 (Fall) **Group Research on Bose–Einstein Condensation**  
CUHK-Shenzhen  
Group leader. Coordinated literature study of “Observation of Bose-Einstein Condensation in a Dilute Atomic Vapor”; delegated tasks, co-authored LaTeX presentation slides, and delivered the class presentation.
- 2025 (Spring) **Group Research on Bell Test in Quantum Physics**  
CUHK-Shenzhen  
Group leader. Coordinated a study of experimental verification of Bell's theorem; delegated tasks, co-authored the final report, and delivered the presentation.
- 2025 (Spring) **Group Research on Computational Physics: Quantum Harmonic Oscillator**  
CUHK-Shenzhen  
Used finite-difference discretization and matrix eigenvalue methods for the one-dimensional time-independent Schrödinger equation. Compared numerical and analytical solutions, examined convergence with grid resolution and domain width, and synthesized team members' results into a final LaTeX report.
- 2024 (Fall) **Group Research on Semiconductor Optical Properties**  
CUHK-Shenzhen  
Group leader. Coordinated a study of semiconductor optical properties; conducted literature review, authored the final report, and delivered the class presentation.

## Teaching and Work Experience

- Oct–Dec 2025 **Undergraduate Student Teaching Fellow (USTF)**  
CUHK-Shenzhen School of Science and Engineering  
Provided tutoring in the Student Help Room, a form of USTF service, for CSC1005 (Introduction to Computer Engineering), CSC1001 (Introduction to Computer Science), and PHY1001 (Mechanics), 23 October–12 December 2025. Help Room certificate available on the portfolio.
- 2025–2026 **Study Abroad Student Ambassador**  
CUHK-Shenzhen Office of Academic Links · Australia and New Zealand Group  
Served as a liaison between students and OAL for communication, safety reminders, and emergency handling; supported students' academic and living experiences, organized group activities, and contributed to outbound-study sharing sessions. Recognized by OAL for exemplary performance.
- 2026 (Spring) **Study Abroad Instagram Ambassador**  
University of Auckland · Auckland, New Zealand  
March–June 2026. Produced multimedia content for the official Study Abroad Instagram account; recognized by the Study Abroad Team as the most prolific ambassador with the highest total likes across posts. Recommendation letter available on the portfolio.
- 2023–2025 **Private Tutor**  
Self-employed  
Tutored CET-4, college entrance English, and international middle-school mathematics using the Big Ideas Math curriculum; tailored lesson plans to students' needs.
- 2023 (Summer) **Teaching Assistant**  
New Oriental Education and Technology Group · Chongqing, China  
Supported English instruction, classroom management, teaching-material preparation, and communication with parents.

## Honours and Awards

- 2023–2026 **Full-Tuition Scholarship**  
CUHK-Shenzhen  
Philip Yu Hong Wong Anita Fung Yee Leung Full-tuition Scholarship. Awarded based on outstanding performance in the 2023 Gaokao.

2026 **Dean's List**  
CUHK-Shenzhen School of Science and Engineering  
Recognized for academic excellence in AY2025–2026. Certificate pending issuance.

2025 **Dean's List**  
CUHK-Shenzhen School of Science and Engineering  
Recognized for academic excellence in AY2024–2025.

2024 **Dean's List**  
CUHK-Shenzhen School of Management and Economics  
Recognized for academic excellence in AY2023–2024.

2024–2025 **Shaw Overcoming Adversity Award**  
CUHK-Shenzhen Shaw College  
Recognized for resilience and perseverance in overcoming personal and academic challenges.

Certification

2025 HCIA-AI: Huawei Certified ICT Associate — Artificial Intelligence. Issued by Huawei Technologies Co., Ltd.; valid through December 2028. Certificate No. 010102001810062372927192.

Skills

|                      |                                                                                                                                                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scientific computing | Python, PyTorch, NumPy, SciPy, Matplotlib, CUDA, Git, LaTeX; GPU experiment workflows through Slurm. Coursework includes CSC1001 (A grade), CSC1006 (machine-learning concepts), and PHY2650 (team project awarded full marks plus an excellence bonus). |
| Machine learning     | Diffusion models, Flow Matching, and Diffusion Transformers (DiT) for AI-for-Science applications.                                                                                                                                                       |
| Collaboration        | Microsoft Word, Excel, PowerPoint, Teams, and Planner.                                                                                                                                                                                                   |
| Content creation     | Instagram, WeChat Official Accounts, and TikTok; photo/video editing with iMovie and TikTok Studio.                                                                                                                                                      |

Languages

|           |                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|
| English   | Full professional proficiency. TOEFL iBT: 100 (Reading 29, Listening 26, Speaking 23, Writing 22), October 2025. |
| Mandarin  | Native proficiency.                                                                                              |
| Cantonese | Basic conversational ability.                                                                                    |

## Extracurricular Activities

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|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Music and the arts | An unwavering love of the arts alongside my studies and research, expressed through choral singing and public performance.                                                                                                                                                                                              |
| 2026 (Fall)        | <b>CUHK-SZ Chorus</b><br>Choir Member<br>Joined the university chorus in Fall 2026, continuing my commitment to choral music.                                                                                                                                                                                           |
| 2026 (Spring)      | <b>Auckland University Student Choir (AUSC)</b><br>Tenor · University of Auckland exchange<br>Performed in two public concerts: the ANZAC Eve Commemorative Concert (24 April 2026) and <i>Oh, the Places We'll Go</i> (29 May 2026). <a href="#">Performance record</a> ; <a href="#">six-video concert playlist</a> . |
| 2025 (Fall)        | <b>School of Music Choir, CUHK-SZ</b><br>Choir Class Member<br>Participated in the choir class and performed in the end-of-term concert, <i>Winter Carols</i> , on 12 December 2025.                                                                                                                                    |
| 2024–Present       | <b>CUHKSZ Economics Club</b><br>Core Management; Head of Secretary Department<br>Managed documentation, social media content, and event logistics.                                                                                                                                                                      |
| 2024–Present       | <b>CUHKSZ Physical Society</b><br>Management<br>Organized academic events and coordinated internal operations.                                                                                                                                                                                                          |
| 2023–2024          | <b>CUHKSZ Shaw College</b><br>Student Helper<br>Created WeChat content, provided multimedia support, and assisted in campus-event organization.                                                                                                                                                                         |
| 2023               | <b>CUHKSZ School of Humanities and Social Sciences</b><br>Research Assistant<br>Supported research documentation through field photography and video editing; assisted professors with data organization.                                                                                                               |